📝 Lecture 8 – Introduction to Machine Learning

MCQs (Dr. Ehab Essa Exam Style)

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1. Machine Learning is best defined as:

a) Writing rules manually
b) Learning without data
c) Learning from experience
d) Hard-coded logic

Answer: C

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2. According to Arthur Samuel, Machine Learning gives computers the ability to:

a) Execute algorithms
b) Learn without explicit programming
c) Store data
d) Improve hardware

Answer: B

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3. In traditional programming, you provide data and:

a) Labels
b) Output
c) Rules
d) Model

Answer: C

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4. In Machine Learning, the system produces the:

a) Data
b) Rules
c) Hardware
d) Compiler

Answer: B

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5. Which learning type uses labeled data?

a) Unsupervised
b) Reinforcement
c) Supervised
d) Clustering

Answer: C

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6. Classification is mainly used to predict:

a) Continuous values
b) Cluster IDs
c) Discrete labels
d) Distances

Answer: C

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7. Regression predicts values that are:

a) Discrete
b) Binary
c) Continuous
d) Random

Answer: C

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8. Unsupervised learning does NOT require:

a) Input data
b) Structure
c) Labels
d) Features

Answer: C

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9. Clustering belongs to which learning type?

a) Supervised
b) Reinforcement
c) Unsupervised
d) Weakly supervised

Answer: C

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10. Reinforcement learning is based mainly on:

a) Labels
b) Rewards
c) Distance
d) Features

Answer: B

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11. In reinforcement learning, the learner is called:

a) Classifier
b) Model
c) Agent
d) Validator

Answer: C

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12. The mapping $f: X \rightarrow Y$ represents:

a) Dataset
b) Target function

c) Feature space
d) Hyperparameter

Answer: B

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13. Training data consists of:

a) Only inputs
b) Only outputs
c) Input–output pairs
d) Rules only

Answer: C

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14. The purpose of feature extraction is to:

a) Reduce training data
b) Encode data meaningfully
c) Generate labels
d) Train faster hardware

Answer: B

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15. A good feature should be:

a) Correlated
b) Random
c) Robust
d) Redundant

Answer: C

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16. Which is NOT a general principle of representation?

a) Robust
b) Discriminative
c) Independent
d) Redundant

Answer: D

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17. The function g(x) learned by the model is called:

a) Target function
b) Hypothesis
c) Dataset
d) Label

Answer: B

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18. Which model class belongs to deep learning?

a) Decision Trees
b) KNN
c) Neural Networks
d) Naive Bayes

Answer: C

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19. KNN classifies data based on:

a) Rules
b) Distance
c) Probability
d) Reward

Answer: B

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20. Manhattan distance is used in:

a) Neural networks
b) Regression
c) KNN
d) Decision trees

Answer: C

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21. Hyperparameters are parameters that:

a) Are learned automatically
b) Change during testing
c) Are fixed before training
d) Are outputs

Answer: C

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22. Choosing value of k in KNN is an example of:

a) Training
b) Labeling
c) Hyperparameter tuning
d) Feature extraction

Answer: C

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23. Which dataset should be used only once at the end?

a) Training set
b) Validation set
c) Test set
d) Feature set

Answer: C

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24. Using the test set for tuning hyperparameters is:

a) Recommended
b) Efficient
c) A bad idea
d) Mandatory

Answer: C

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25. Validation set is mainly used to:

a) Train model
b) Tune hyperparameters
c) Generate labels
d) Store features

Answer: B

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26. Cross-validation is useful when:

a) Dataset is large
b) Dataset is small
c) Labels are missing
d) Features are noisy

Answer: B

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27. In k-fold cross-validation, the data is:

a) Deleted
b) Duplicated
c) Split into folds
d) Randomized once

Answer: C

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28. Image categorization is an example of:

a) Regression
b) Clustering
c) Classification
d) Reinforcement

Answer: C

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29. CIFAR-10 dataset contains:

a) 2 classes
b) 5 classes
c) 10 classes
d) 100 classes

Answer: C

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30. The main goal of Machine Learning is to:

a) Memorize data
b) Learn rules from data
c) Replace programming
d) Eliminate testing

Answer: B